

Deep Learning for Perception – Research Project

Objective

This project trains students to conduct end-to-end research in computer vision. Each student (or group) will identify a problem, design a solution using deep learning methods, evaluate results, and document findings in the form of a research paper. The process will mimic real academic research and prepare students for conference-level submissions.

Scope

Projects must be aligned with course topics:

- CNNs and CNN architectures
- Image classification
- Segmentation (semantic, instance, etc.)
- Object detection
- Optical Character Recognition (OCR)
- Vision Transformers (ViTs)
- Generative modeling (GANs and variants)
- Multimodal perception

Process and Deliverables

The project runs in **phases**. At each phase, both the research progress and the corresponding section of the paper must be submitted in the provided IEEE LaTeX template.

- 1. Phase 1: Problem Identification and Research Planning**
 - Identify a problem in the domain of perception.
 - Search and select a suitable dataset.
 - Submit: Introduction draft (problem domain, importance).
- 2. Phase 2: Literature Review and Problem Statement**
 - Survey related work.
 - Define research gap, problem statement, and research questions.
 - Submit: Related Work and Problem Statement sections.
- 3. Phase 3: Methodology and Experiment Design**
 - Propose methodology and model architecture.
 - Define evaluation metrics.
 - Submit: Methodology section with initial experimental setup.
- 4. Phase 4: Experiments and Results**
 - Run experiments on chosen dataset.
 - Collect and analyze results.
 - Submit: Results section (tables, figures). Code (Jupyter Notebook)
- 5. Phase 5: Discussion and Final Paper**
 - Discuss findings, limitations, and future directions.
 - Submit: Discussion and Conclusion sections.
 - Submit final complete research paper in IEEE LaTeX format.

Evaluation

- Quality of research problem and dataset selection.
- Depth and clarity of literature review.
- Soundness of methodology and experiments.
- Quality of results and discussion.
- Completeness and writing quality of research paper.

Tools and Resources

- IEEE LaTeX template will be shared.
- Students may use PyTorch, TensorFlow, or Keras for implementation.
- Students are encouraged to use public datasets of good quality.

Upcoming Conferences

Name	Start Date	Location	External URL
2026 IEEE International Conference on Data Science, Machine Learning, and Business Intelligence	3/26/2026	Lahore, Pakistan	https://datasci.world/
1st International Conference on Computing Sciences and Emerging Trends	2/11/2026	Nawabshah, Pakistan	https://cset.quest.edu.pk/home
INTERNATIONAL CONFERENCE ON BIOLOGICAL RESEARCH AND APPLIED SCIENCE	1/20/2026	Karachi, Pakistan	https://ibras.juw.edu.pk/
International Conference on Innovations in Information and Communication Technologies	1/15/2026	Jamshoro, Pakistan	https://ic-iict.muet.edu.pk/
8th International Multi-topic ICT Conference 2025	12/29/2025	Jamshoro, Pakistan	https://imtic.muet.edu.pk/
8th International Multi-Topic ICT Conference 2025	12/29/2025	Jamshoro, Pakistan	http://imtic.muet.edu.pk/
27th International Conference on Multi Topic (INMIC 2025)	12/23/2025	Islamabad, Pakistan	https://inmic2025.riphah.edu.pk
3rd International Conference on Recent Advances in Computing, Artificial Intelligence, and Data Science	12/23/2025	Islamabad, Pakistan	https://www.caids.org.pk/index.php

5th International Conference on Digital Futures and Transformative Technologies	12/17/2025	Islamabad, Pakistan	https://icodt2.sines.nu.st.edu.pk/
Horizons of Information Technology and Engineering (HITE) 2025	12/11/2025	Lahore, Pakistan	https://hite.ucp.edu.pk/
2025 6th International Conference on Innovative Computing (ICIC)	12/10/2025	Lahore, Pakistan	https://icic.umat.edu.pk/
15th International Conference on Mathematics, Actuarial, Computer Science, Statistics	12/6/2025	Karachi, Pakistan	https://macs.iobm.edu.pk/
4th International Conference on Andalusian Literature: The Cultural & Artistic Interaction between Arabs and West	11/25/2025	Islamabad, Pakistan	https://andalusianlit.aiou.edu.pk/
Biennial Conference of Comparative Education Society of Asia International Conference	11/24/2025	Islamabad, Pakistan	https://cesa2025.aiou.edu.pk/
20th International Conference on Emerging Technologies (ICET-25)	11/18/2025	Peshawar, Pakistan	https://www.uetpeshawar.edu.pk/icet/
3rd International Conference Advancing Research in Social Sciences	11/17/2025	Islamabad, Pakistan	https://icarss2025.aiou.edu.pk/
International Conference on Engineering & Computing Technologies 2025	11/13/2025	Islamabad, Pakistan	https://numl.edu.pk/icet/
International Conference on Cybersecurity and Digital Defense	10/24/2025	Karachi, Pakistan	https://cs.juw.edu.pk/CyDD/
X. ASC 2025 / Fall	10/16/2025	Faisalabad, Pakistan	https://asc.academicianstudies.com/en/
4th International Conference on Communication, Computing and Digital Systems	10/1/2025	Islamabad, Pakistan	https://ccode.bahria.edu.pk/